INCH-POUND

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MILITARY SPECIFICATION

TRUCK SERVICING, 3 TON, 27 FOOT ELEVATION FOR LOADING C-5 AIRCRAFT UPPER DECK

This specification is approved for use within WR-ALC, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a truck to service C-5 aircraft with a door threshold at 27 feet above ground level. The truck transport payload shall be at least 3 tons.

1.2 Classification. The vehicle shall employ a mechanism to lift the van body to the required height.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

W-B-131	Battery, Storage: Vehicular, Ignition, Lighting and Starting
VV-F-800	Fuel Oil, Diesel

Military

MIL-P-514	Plates, Identification, Instructions and Marking, Blank
MIL-T-83133	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8
MIL-H-83282	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Aircraft

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMVRG, Robins AFB, GA 31098-5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 2320

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STANDARDS

Federal

FED-STD-595	Colors
FED-STD-297	Rustproofing of Commercial (Nontactical) Vehicles

Military

MIL-STD-889	Dissimilar Metals
MIL-STD-1223	Nontactical Wheeled Vehicles Treatment, Painting, Rustproofing, Identification Marking and Data Plate Standards
MIL-STD-1791	Designing for Internal Aerial Delivery in Fixed Wing Aircraft
MS-51317	Light, Warning, Vehicular, Rotating
ZZ-T-381	Tires, Pneumatic, Vehicular (Highway)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Department of Transportation (DOT)

Federal Motor Carrier Safety Regulations

Federal Motor Vehicle Safety Standards

(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington DC 20591.)

Environmental Protection Agency (EPA)

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines

Noise Emission Standards for Transportation Equipment – Medium and Heavy Trucks

(Application for copies should reference the code of Federal Regulations 40CFR and the Federal Register and should be addressed to the Superintendent of Documents, US Government Printing Office, Washington DC 20402.)

Design Handbooks

AFSC DH 1-11	Air Transportability

(Copies of this design handbook should be obtained from HQ AFLC, ASD/ENECA, Wright Patterson AFB OH.)

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Drawings

U.S. Air Force drawing number 8838302

(Copies of this drawing should be obtained from WR-ALC/MMVRG, Robins AFB GA 31098.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

SAE, Incorporated

SAE Standards and Recommended Practices

ARP 1801	Exterior Sound Level of Specialized Aircraft Ground Support Equipment, Measurement of
J537	Storage Batteries (DOD adopted)
J551	Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (20-1, 000 MHZ)
J588	Turn Signal Lamps
J589	Turn Signal Switch
J682	Rear Wheel Splash and Stone Throw Protection (DOD adopted)
J683	Tire Chain Clearance – Trucks, Buses and Combinations of Vehicles (DOD adopted)
J688	Truck Ability Predication Procedure (DOD adopted)
J1292	Automobile, Truck, Truck-Tractor, Trailer and Motor Coach Wiring
J1349	Engine Power Test Code – Spark Ignition and Diesel

(Application for copies should be addressed to SAE, INCORPORATED, 400 Commonwealth Drive, Warrendale PA 15096.)

The Tire and Rim Association, Incorporated

Yearbook

(Application for copies should be addressed to the Tire and Rim Association, Incorporated, 3200 W Market Street, Akron OH 44313.)

US Department of Agriculture (USDA)

5100-1A	Spark Arrestors for Internal Combustion Engines

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(Application for copies should be addressed to US Department of Agriculture, Forest Service, Equipment Development Center, Dan Dimas CA 91773.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.12, the vehicle, components, assemblies and accessories to be delivered under the contract shall be standard or optional items which meet or exceed the requirements of this specification. All chassis items shall be as represented and rated in the chassis manufacturer's technical data, and special bodies or mounted equipment shall be as represented and rated in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be on file in the engineering offices of the procuring activity prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of the invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

3.1.1.1 Painting and marking. Vehicle treatment, painting and data plates shall be in accordance with MIL-STD-1223. The exterior color shall be semigloss green, matching Color Chip Number 24052 of FED-STD-595. The complete exterior of the body shall match the exterior color of the vehicle. Identification markings shall be in accordance with the Air Force requirements of MIL-STD-1223.

3.1.1.1.1 Caution plates. Caution plates shall be furnished and mounted in conspicuous locations in the driver's compartment and at the operator's control stations. Plates shall be in accordance with Type I, Style 4 or Type III, composition A or C of MIL-P-514.

3.1.1.1.2 Instruction plate. A photo-etched corrosion resistant plate, in accordance with Type II, composition C of MIL-P-514, showing a schematic of all valves, valve controls, equipment and complete instructions for operating the vehicle shall be fastened at the operator's control station and in the vehicle cab in an accessible location.

3.1.1.2 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297.

3.1.1.3 Drain plugs. Drain plugs installed in rear axles shall be of the permanent magnet type.

3.1.1.4 Heavy duty cooling system. A heavy duty cooling system shall be furnished that will maintain engine coolant at a temperature below the boiling point with vehicles loaded to rated Gross Vehicle Weight (GVW) and operated at an altitude of 10,000 feet above sea level or in an ambient temperature of not less than 125°F.

3.1.1.5 Coolant system indicators. The coolant system shall include a deaeration system, a surge tank, or a coolant recovery reservoir of not less than 2 quart capacity. A high coolant temperature or low coolant level alarm buzzer and red indicator warning light shall be provided on the dash instrument panel.

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3.1.1.6 Wood treatment. Wood surfaces shall be treated in accordance with MIL-STD-1223.

3.1.1.7 Towing devices. Not less than two hooks, loops, or pins for towing the vehicle shall be furnished on the front of the vehicle, and not less than two hooks, loops or pins shall be furnished on the rear of the vehicle.

3.1.1.8 Rear wheel splash and stone throw protection. Rear wheels shall have rigid (constructed of metal or rubber with metal trim) splash shields in the front of the rear wheels and rubber mud flaps at the rear of the rear wheels. Removable splash shields and mud flaps shall be fastened to the chassis. When tilt cabs are furnished, rubber mud flaps shall be provided to the rear of the front wheels. Splash and stone throw protection shall be in accordance with SAE J682. A metal strip, not less than 1/8 inch thick and not less than one inch wide extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flaps. Mud flaps shall be an all black color.

3.1.1.9 Spark arrestor. As specified herein (see 3.4.4), the vehicle shall be furnished with a spark arrestor.

3.1.1.10 Operating temperature. The engine shall be capable of starting in any ambient temperature from -20°F to +125°F, and the vehicle shall be capable of operating in an ambient temperature from -40°F to +125°F. In addition, the specified components shall be capable of meeting the ambient temperatures specified herein.

3.1.1.11 Tiedown devices. The vehicle shall be provided with tiedown devices to permit tiedown of the vehicle to the floor or deck of a transportation medium including the C-5 Aircraft. Tiedown devices shall prevent shifting or movement in any direction. The tiedown devices shall be in accordance with Class 2, Type III of MIL-STD-209. Tiedown devices shall be identified by an information plate conforming to Type I, Style 4 of MIL-P-514. The plate shall be attached to a flat surface and shall not extend beyond the edges of that surface. Markings shall clearly indicate that the device is intended for tiedown of the vehicle.

3.1.1.12 Fasteners. All fasteners of Grade 5 or higher shall meet the quality assurance requirements of U.S. Air Force drawing number 8838302.

3.2 Design requirements.

3.2.1 Federal motor vehicle safety standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with the Environmental Protection Agency (EPA) regulations governing control of air pollution from new motor vehicles and new motor vehicle engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with the state of California regulations governing air pollution control in effect on the date of manufacture.

3.2.3 Sound level. The maximum sound level produced by the vehicle shall not exceed 84dBA when measured in accordance with SAE ARP 1801. An additional 2 dBA allowance shall be permitted for production units to provide for variation in test site, temperature gradients, test equipment, and inherent differences in nominally identical vehicles.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis, cab and body, with all attachments, accessories and equipment and a full complement of fuel, lubricants and coolant.

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3.2.5 Gross vehicle weight (GVW). The minimum GVW shall consist of curb weight, operator and one passenger (weight computed at 200 pounds each) and a 6,000 pound payload, distributed uniformly over the load area, to provide not less than the minimum GVW required in the vehicle design. In addition, the actual GVW rating of the vehicle furnished shall exceed the minimum GVW required.

3.2.6 Capacities. The suspension, axles and tires shall have a rated capacity at least equal to 1.1 times the load imposed on each component.

3.2.7 Ratings. Vehicle ratings shall be manufacturer's published rating. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be submitted to the engineering office of the procuring activity. The ratings shall be established by the manufacturer of the equipment being verified. Ratings shall be submitted with the questionnaire (commercial vehicle engineering data).

3.2.8 Air transportability. The vehicle shall be air transportable in C-5 Aircraft in accordance with the requirements of MIL-STD-1791 and Design Handbook AFSC DH 1-11. Removal or relocation of mechanically attached (nonwelded, nonriveted, etc) components with common hand tools, requiring not more than one manhour total to remove, relocate and tiedown and not more than one manhour total to return the vehicle to its original, as opposed to reduced, configuration shall be acceptable. The self mobility of the vehicle shall not be affected. Tiedowns for removed or relocated equipment shall be furnished. Removal, relocation, tiedown and installation instructions shall be described in the equipment manual(s) delivered. The vehicle, in its reduced configuration, shall not exceed any of the dimensions in DH 1-11. The curb weight of the vehicle shall not exceed 10,000 pounds on the front axles and 20,000 pounds on the rear bogie. In addition to the requirements of 3.2.6, the rated capacity of the axles and suspension system shall be not less than 1 1/4 times the load imposed on each by the curb weight of the vehicle. The vehicle shall be air transportable as described above without any other special provisions and without any shoring. The vehicle shall not be delivered to the Government in its reduced configuration.

3.2.9 Air transportability verification. Comprehensive scale drawings, satisfactorily showing each stage of loading, tiedown and unloading for the aircraft specified above, shall be furnished to the engineering offices of the procuring activity, TACOM, AMSTA-UED, for subsequent review and approval of responsible Air Force officials at Headquarters, Aeronautical Systems Division (AFSC) Wright Patterson AFB. AFSC Design Handbook DH 1-11, entitled "Air Transportability Handbook", is recommended for guidance and verification of air transportability and may be obtained from ASD/ENESS Wright Patterson AFB OH 45433, telephone (513) 255-6281.

NOTE

Government approval of air transportability will be required before final acceptance of any vehicles.

3.2.10 Overall width. The overall width of the vehicle, exclusive of mirrors, lights, reflectors and tires shall be not more than 96 inches. Width over tires shall be not more than 102 inches.

3.2.11 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

3.2.12 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the trucks may be newly fabricated from recovered materials to the maximum extent practicable, provided the trucks produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the trucks.

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3.3 Performance.

3.3.1 Speeds and gradeability. The vehicle loaded to its specified GVW (see 3.2.5) shall ascend 1% continuous grades at 50 miles per hour (MPH). Gradeability requirements shall be met with the main transmission in direct drive and, when a multispeed axle or auxiliary transmission is furnished, with the axle or auxiliary transmission in highest speed ratio. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

3.3.1.1 Low speed. Low speed shall be calculated with the engine operating at not less than 35% of the recommended governed speed, and shall provide not more than 5 MPH vehicle speed.

3.3.1.2 Maximum geared speed. Maximum geared speed at engine governed speed shall be not less than 58 MPH. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum Geared Speed} = \frac{\text{Governed Speed (RPM)}}{\text{Total Gear Reduction X Tire Factor (see 6.3)}}$$

3.3.2 Service brakes. Service brakes shall control and hold the vehicle when the vehicle is loaded to its specified GVW (see 3.2.5) on a 30 percent grade. The service brakes shall stop the vehicle loaded to specified GVW (see 3.2.5) within the stopping distance requirements of Federal Motor Carrier Safety Regulation (FMCSR) 393.52 for property carrying vehicles with a GVWR more than 10,000 pounds.

3.3.2.1 Parking brake. The parking brake shall be completely independent from the service brakes and shall be capable of holding the truck loaded to its specified GVW (see 3.2.5) on a 15 percent grade, both up and down the slope, with no movement of the truck. The parking brake shall be in accordance with FMCSR 393.41.

3.3.3 Subframe, hoist and body performance. In addition to the requirements in 3.3.3.1 through 3.3.8, the subframe, hoist and body shall be designed to meet the static and proof tests specified in Section 4 of this specification.

3.3.3.1 Stability. Stability shall be demonstrated with and without payload. The body shall be level, stable and rigid throughout the entire elevation and retraction of the hoist mechanism. In the fully elevated position, the body floor shall not deflect more than 1/2 inch, measured from the floor to the top of the subframe. The vehicle in any elevated position shall show no evidence of the tires or outriggers departing from the ground.

3.3.3.2 Lifting capacity. The rated lifting capacity of the hoist shall be not less than the sum of the weight of the body platform and ramp plus two times the specified payload.

3.3.3.2.1 Height. The hoist shall be capable of raising the body and payload from rest position to a minimum height of 27 feet at the forward edge of the extended ramp measured from the ground, with the body level.

3.3.3.3 Manual locking device. A manual locking device shall be provided to retain the body and hoist at a raised height of not less than 96 inches measured from the ground, with the body level.

3.3.3.4 Operating speeds. The hoist and body shall be capable of elevating and retracting with and without the specified payload (see 3.2.5) at not less than 15 feet per minute (FPM) and not more than 20 FPM, with the vehicle engine at idle speed and when the load is imposed on the hydraulic system.

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3.3.3.5 Position stability. When the vehicle is on level ground, the hoist hydraulic system shall hold the body with the specified payload in any elevated position without more than 1/2 inch vertical settling in 30 minutes.

3.3.3.6 Unbalanced load. With 1/4 of the specified payload (see 3.2.5) located in any corner of the body floor, the hoist shall be capable of raising and lowering the body without evidence of binding, deformation or bending of the hoist mechanism. The body shall not deflect more than 1/2 inch, as specified in 3.3.3.1. The body shall remain level during the raising and lowering.

3.3.3.7 Wind resistance. The vehicle, loaded to specified payload (see 3.2.5) and unloaded, and at the fully elevated position, shall be capable of withstanding 35 MPH winds and wind gusts up to 70 MPH from any direction without overturning and without permanent deformation of any component. Wind load shall be calculated with the maximum side area perpendicular to the wind force and shall be measured in accordance with the following:

a. At the body, when the body is fully retracted.
b. At the body and hoisting mechanism, when the body is elevated to its maximum specified height, in the most critical position that would induce tipping.

In both a and b, the resistant moment shall be greater than the wind load. The contractor shall furnish calculations or a certified test report for the vehicle or an identical group by type, class, model and payload to the engineering office of the procuring activity. Hydraulic extended outriggers shall be used to achieve assistance moment.

3.3.3.8 Stress analysis. A stress analysis of the vehicle body and frame shall be furnished as specified in 3.7.

3.4 Chassis components.

3.4.1 Engine. The engine furnished for the specified vehicle class shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

3.4.1.1 Diesel. The vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine, with not less than four cylinders. Engine net horsepower figures used in performance prediction calculations shall be determined in accordance with SAE J1349. A fan clutch to reduce fan speed automatically, when not required for engine cooling, shall be furnished. The engine shall be capable of providing the performance specified herein when operating on diesel fuel conforming to VV-F-800, and with a reduction in performance of not more than 20 percent when operating on Grade JP-8 turbine fuel conforming to MIL-T-83133. Engine manufacturer's certification is acceptable.

3.4.1.2 Oil filter. A full flow type engine oil filter shall be furnished.

3.4.1.3 Governor. An engine governor shall be furnished, and set and sealed to limit the engine to the engine manufacturer's maximum recommended operating speed.

3.4.1.4 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. The control shall include partial thermostatic control of all coolant flow through the radiator shutters, or complete thermostatic control of all coolant flow through the radiator.

3.4.1.5 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater, engine oil heater, battery heater, hydraulic fluid reservoir heater and a fuel warmer shall be provided. The heaters shall operate on 110 volt alternating current (AC) and shall be wired through a junction block to a single three pronged (male), weatherproof, slave receptacle for receiving external power and grounding the

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vehicle. A three wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard weatherproof, three pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation (393.77(c)(7)). Electrical insulation of the connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to -60°F without cracking or loss of dielectric capacity). All heater lead wires shall be installed without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters shall be furnished as follows:

a. Coolant heater, 1500 watt minimum rating, shall be installed in the engine block or lower, coolant inlet hose. An engine thermostat with an operating range of 170°F to 195°F shall be installed.
b. Immersion type engine oil heaters, 300 watt minimum rating, with 170°F to 195°F thermostat, shall be installed in the oil pan through any convenient opening.
c. Battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air at -60°F thermostat to limit the temperature of the electrolyte to not more than 80°F.
d. Hydraulic fluid reservoir heater shall have a capacity to maintain the hydraulic fluid at a temperature of not less than 10°F during vehicle exposure in ambient temperature as low as -60°F and shall embody a thermostat to limit the temperature of the hydraulic fluid to not more than 75°F.
e. A fuel warmer or preheater shall be provided to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of -40°F to an outlet temperature of +9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

3.4.1.6 Silicone rubber hoses. Silicone rubber radiator and heater hoses shall be furnished. Hose clamps suitable for silicone rubber hoses shall be used.

3.4.2 Electrical systems. Electrical systems shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

3.4.2.1 Starting system. 12 or 24 volt starting system, with 12 volt direct current (DC) lighting system and not less than 60 ampere alternator shall be furnished. Engine starting equipment shall include a measured shot type ether starting system or glow plug. If an ether system is furnished in lieu of a glow plug, it shall be of the measured shot type. Measured shot type ether system shall be key operated or manually operated from the driver's compartment and shall be inoperative with the engine warm. A replaceable ether reservoir of not less than 12 fluid ounces shall be furnished.

3.4.2.2 Battery. Each battery shall be of 12 volt potential. The total reserve capacity rating and total cold cranking ampere rating at 0°F, both measured in accordance with SAE J537, shall be not less than that specified in Table I. Batteries shall be of the maintenance free type having the maintenance free characteristics listed in W-B-131.

TABLE I BATTERIES

Engine Type	Reserve Capacity (minutes)	Cold Cranking (amperes)
Diesel (less than 220 gross horsepower)	320	1200
Diesel (220 gross horsepower and over)	480	1800

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3.4.2.3 Lighting. All vehicle lights and reflectors shall be as specified herein and shall be in accordance with Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20 and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on vertical surfaces of the rub rails (unless recessed and fully protected) or on vehicle bumpers. Rear identification lamps, marker lamps, and reflectors shall be mounted on the body.

3.4.2.4 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, Class A and shall be mounted on the steering column. Vehicles shall be provided with double faced front signal units and with single faced rear signal units installed in accordance with SAE J588. Turn signals shall not be mounted on the engine compartment hood. Turn signal operating units shall have visible and audible flash indicators.

3.4.2.5 Spotlights. Two sealed beam spotlights of not less than 4 1/2 inch diameter and 100 watts shall be furnished. The spotlights shall be mounted on each side of the vehicle cab and shall be operable from within the vehicle cab. The spotlights shall be capable of being adjusted from within the vehicle cab.

3.4.2.6 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

3.4.3.1 Air cleaner. An air cleaner shall be furnished. When equipped with an oil-bath type, oil capacity shall be not less than one quart. Air cleaner shall be the flame arresting type.

3.4.3.2 Fuel tank(s). Fuel tank(s) shall be not less than 50 gallons total capacity. When more than one tank is furnished, means shall be provided to assure equalized fuel levels in both tanks.

3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. Horizontal exhaust system shall be furnished. Except for turbocharged engines, a spark arrestor shall be furnished in accordance with USDA Standard 5100-1, and equipped with a cleanout plug of at least 3/4 inch diameter.

3.4.4.1 Exhaust system safety shield. The exhaust system or hydraulic system shall be shielded so that, in the event of a hydraulic line rupture or hydraulic fluid leak, the hydraulic fluid will not come into contact with any part of the vehicle exhaust system.

3.4.5 Gear train.

3.4.5.1 Automatic transmission. The vehicle shall be provided with an automatic transmission. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission shall be of the fixed ratio type and shall provide continuous drive. The transmission shall include a hydraulic torque converter and not less than four forward gear ratios without movement of the selector. A drive gear for the power takeoff shall be provided conducive to the configuration and maximum performance requirements specified herein. Hydraulic lines and other associated parts which are fastened to the assembly shall be positioned for easy inspection, ready maintenance and possible replacements and shall have maximum protection against damage.

3.4.5.2 Power take-off. The power take-off shall be of a rated capacity to operate powered equipment. Controls to operate the power take-off shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKE-OFF ENGAGED" shall be provided and installed so as to be readily visible to the seated driver.

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3.4.6 Driveline components. Driveline components shall be adequate to transmit the maximum delivered torque to the engine, as developed through the maximum gear train reduction.

3.4.7 Frame. The chassis frame shall be the manufacturer's standard for the type and class vehicle furnished. The vehicle shall have a heavy duty main frame or frame with reinforcements. A heavy duty frame shall have frame rails of a greater section modulus than the manufacturer's standard for the class vehicle furnished and shall provide structural strength at least equivalent to the reinforced frame specified herein for the type vehicle furnished. Reinforcements shall extend at least from the rear of the front suspension, rear hanger bracket to the front of the rear spring, front hanger bracket. Reinforcement shall provide sufficient structural strength in the chassis frame through increased resisting bending moment to at least equal the load imposed, with body loaded and in any elevated position, to the specified GVW. Chassis frame rails shall not project beyond the rear end of the body. There shall be no cracking of the frame during the life of the truck when used as specified in this document.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system with components having a rated capacity of at least equal to the load imposed on each member, measured at the ground, with the vehicle loaded to specified GVW. When suspension capacity is rated at the spring pads, unsprung weight shall be deducted. The vehicle shall be equipped with hydraulic double acting shock absorbers at the front wheels. Suspension system ratings shall include the loads imposed during all operating conditions.

3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with vehicle loaded to specified GVW and during all operating conditions. Unless a two speed axle is specified, a single reduction rear axle shall be furnished.

3.4.10 Wheels, rims, tires and tubes. The vehicle shall be equipped with single front and dual rear wheels. Except as specified herein, rims and tire ratings shall conform to Tire and Rim Association recommendations for the type and size of tires furnished. Rim and wheel sizes shall be the same for all wheels on each vehicle. Tire size and load range (ply rating) shall be the same for all tires on each vehicle.

3.4.10.1 Tires. Tires shall be tube or tubeless type, with highway tread. Steel belted radial tires shall be furnished. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with vehicle loaded to specified GVW. Tires shall conform to the Tire and Rim Association recommendations, or to ZZ-T-381 with a size designation system the same as the Tire and Rim Association.

3.4.10.2 Inner tubes. When tube type tires are furnished, inner tubes shall be of heavy duty type, and shall be of proper size for tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

3.4.10.3 Tire chain clearance. Tire chain clearance shall be furnished. Clearance shall be in accordance with SAE J683.

3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43 and 393.45 through 393.52.

3.4.11.1 Service brakes. The vehicle shall be equipped with full air brakes on all wheels. The braking system, complete with all necessary components, shall include the following:

a. Air compressor, unloader head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 12 CFM.
b. Air storage reservoir(s), with not less than 2,000 cubic inches total capacity each tank, equipped with drain cock and with safety and check valves between compressor and last reservoir tank.
c. Foot control, suspended on treadle type.

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d. Air control valves.
e. Air pressure gage, visible to the driver.
f. Low air pressure warning, visible and audible.
g. Automatic moisture ejector.
h. Service brake stop lamp switch.
i. A dessicant type air dryer shall be furnished.
j. Automatic slack adjusters.

3.4.11.1.1 Brake controls for use from a towing vehicle. The vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, relay emergency valve with no-bleed-back feature (except when spring applied emergency brake is furnished), additional air lines and fittings. The service and emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

3.4.12 Cab. The manufacturer may furnish any standard configuration full-width model cab. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage and with at least the curbside door equipped with external, key-operated lock. Drip rails shall be installed above the cab doors. The cab shall have upholstered, full width adjustable seat and back, or individual adjustable driver's seat, and an individual passenger seat shall be provided. The color of the upholstery and the interior finish shall be compatible with the exterior colors (see 3.1.1.1). White upholstery shall not be furnished. Interior lighting shall be provided. Manufacturer's standard fastenings and two pairs of seat belts shall be installed on the vehicle. Safety grips shall be provided on each side of the cab to assist personnel in entering and leaving the cab.

3.4.13 Steering. Power steering will be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type operated by either air or electric motor(s).

3.4.15 Bumpers.

3.4.15.1 Rear end protection. The rear end of the vehicle shall be protected in accordance with Federal Motor Carrier Safety Regulation 393.86. A rear bumper shall be furnished (see 3.4.25).

3.4.16 Stowage space. Stowage space of sufficient size to accommodate a vehicle jack, hand tools, antiskid chains and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. Storage space for these tools may be furnished inside the cab. When stowage space for these tools is located outside of the cab, it shall be weatherproof and shall provide for locking with a padlock.

3.4.16.1 Tools. Each vehicle shall be furnished with tools required for exchanging mounted tire assembly with a spare assembly, and shall include at least a hydraulic jack, jack handle and wheel nut wrench. The jack shall be of such closed height as to permit its location under the axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of loaded vehicle to a height adequate to permit removal and replacement of wheel and tire assembly.

3.4.17 Heater and defroster. The vehicle shall be provided with a hot water heater with fresh air intakes and discharge outlets to the floor and to windshield defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

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3.4.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. All instruments shall be visible to the driver when seated in the driving position.

3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

a. Key-operated ignition switch.
b. Ammeter.
c. Fuel gauge.
d. Oil pressure gauge (warning light by itself is not acceptable).
e. Engine temperature gauge (warning light by itself is not acceptable).
f. Speedometer with recording odometer.
g. Ash receptacle.
h. Dual sunvisors.
i. Front door mounted armrest on driver's side.
j. Driver's compartment ventilator other than window.
k. Tachometer.
l. Hourmeter (see 3.4.23).

3.4.20 Rearview mirrors. Outside rearview mirrors shall be mounted on each side of the cab. The mirrors shall be of the combination type having flat and convex areas enclosed in a common housing. The flat portion shall have not less than 50 square inches of reflective area. The convex portion shall have not less than 20 square inches of reflective area. The mirrors on conventional cabs shall have not less than three supporting arms. On tilt cabs, the mirrors shall have not less than two supports.

3.4.21 Horn. Manufacturer's standard electric horn shall be furnished.

3.4.22 Fire extinguisher. Two 2 1/2 pound (minimum) Halon 1211 fire extinguishers with mounting brackets shall be furnished, but not installed.

3.4.23 Engine hourmeter. An engine hourmeter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to ensure trouble free performance under severe operating conditions. The engine hourmeter shall be mounted on the vehicle cab instrument panel or in the engine compartment, in readable location.

3.4.24 Back up alarm. A back up alarm shall be provided with an audible pulsating signaling device (electrical or mechanical) to caution personnel when the vehicle is in reverse gear operation.

3.4.25 Rear bumper. A rear bumper extending aft of the body shall be installed on the frame. A safety tread step, integrated with the rear bumper, shall be provided to assist personnel into the body when the body is in the lowered position. The step shall extend across the width of the body and shall be at least 10 inches deep.

3.5 Body.

3.5.1 Construction. The subframe and body shall be designed, engineered and constructed of materials that display the minimum weight conducive to the configuration and maximum performance requirements specified herein. Hydraulic lines and other associated parts which are fastened to the assembly shall be positioned for easy inspection, ready maintenance and possible replacement, and shall have maximum protection against damage afforded by the configuration of the subframe, hoist and body

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assembly. The body shall be attached to the subframe. Rotating bearing surfaces, which are subject to continuous wear, shall be provided with replaceable bearings and bushings. Lubrication fittings shall be provided on all rotating parts. Bearing surfaces shall be adequate to resist torsional moment when inherent with the design.

3.5.1.1 Subframe and body components. The subframe and body structural components shall have a calculated design stress such that the yield point of the materials used shall not be exceeded by an induced stress caused by the weight of the components, plus the stress due to three times the specified payload imposed on the body floor when the vehicle body is in any of its attainable elevated positions. The materials used shall meet all applicable performance requirements specified herein. Alignment shall be maintained under all operating conditions, without applying stress to any attachment. Hoist subframe longitudinal members shall be of continuous length without splices, for the length of the subframe.

3.5.1.2 Subframe and hoist. The vehicle shall be equipped with a subframe that is mounted to the vehicle chassis frame. The subframe shall be the length of the body. The subframe shall consist of the structural members, hydraulic components and cylinder(s).

3.5.1.3 Hoist. The hoist shall be provided to elevate the body. Cylinder(s) and rods shall have length to diameter ratio for forward and lateral stability. Means shall be provided to prevent body shifting when the body is retracted during highway travel.

3.5.2 Subframe mounting. The subframe shall be secured with U-bolts, twin studs, or J-bolts and shall include a wood or steel breaker strip.

3.5.2.1 U-bolts or twin studs. When U-bolts or twin studs are used, there shall be not less than four U-bolts or twin studs per side, having 9/16 inch minimum body diameter with 5/8 inch minimum thread diameter. Tie plates shall be at least 1/2 inch thick. The vehicle chassis frame shall be braced, using wood blocks at each mounting point, unless the mounting point is located at full depth frame crossmember. Blocks shall incorporate a keeper strap or groove for mounting bolt and shall be of a width to assure retention. Two shear bolts shall be provided, one bolted to each side of the rear portion of the subframe, to maintain alignment on the vehicle chassis. Forward mounting bolts shall be located to the rear of the tapered portion of the breaker strips.

3.5.2.2 Brackets. When brackets are used, they shall be bolted to the web of the frame rails. The mounting brackets shall provide a means of drawing down the subframe on the chassis rails. Provisions shall be made to prevent lateral shifting of the breaker strips. When additional holes are required to secure mounting brackets to the chassis frame rails, they shall be located within the area of the rail which is designated as being safe for drilling in accordance with the chassis manufacturer's body builders layouts. Attachments shall not interfere with, nor obstruct existing chassis components. Spring loaded hold down devices may be used to secure the chassis to the subframe and to provide independent action between the chassis and the subframe.

3.5.2.3 J-bolts. When J-bolts are used, one J-bolt shall be located near each point where a body crossmember intersects the vertical plane of the chassis frame rails, to the rear of the tapered section of the breaker strips (see 3.5.2.4). J-bolts shall be not less than 3/8 inch in diameter. J-bolts shall encircle the lower chassis frame flange and shall be securely attached to the body. Two keeper straps, as specified in 3.5.2.1, shall be provided. Provisions shall be made to prevent lateral shifting of the breaker strips.

3.5.2.4 Breaker strips. Hardwood or dense southern yellow pine breaker strips of not less than 1 1/6 inches finished thickness shall be provided. The breaker strip shall be installed between the longitudinal sills and the vehicle chassis and shall have a taper of 1 inch in 18 inches at the forward end.

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3.5.3 Fenders. Rear fenders with heavy rubber splash guards shall be provided over the rear wheels. Fenders shall be of not less than 14 gauge (0.0747 inch) sheet steel. A fender flange around the wheel cutout shall be furnished. Fenders shall be at least the same width as the dual tires. Fenders shall be supported by the chassis frame and shall be demountable.

3.5.4 Electrical wiring. All wiring shall be in conduit. Splices are not acceptable. All wiring ends shall have staked lugs or connectors. Wiring shall be in accordance with SAE J1292. Wiring shall be coded. Floodlights and interior body lights shall be furnished with three way switches at the lamps and on the vehicle chassis cab instrument panel.

3.5.5 Lighting.

3.5.5.1 Floodlights. Two sealed beam, 12 volt, 100 watt, universal mounted floodlights shall be furnished for night cargo transfer operations. Floodlights shall be a minimum of 5 1/2 inches in diameter and shall be corrosion resistant. Floodlights shall be located on the forward roof of the van.

3.5.5.2 Interior lighting. Four interior lights shall be furnished. Two lamps shall be on each side of the body. The lighting shall be recessed within the walls. The interior lights shall provide an average illumination of not less than 40 foot candles, measured throughout the van body at a height of 40 inches above the floor. Lamps shall have a guard to prevent damage. Guards shall be hinged at the top and secured to prevent movement.

3.5.5.3 Warning lights. Two red flashing warning lights and an audible beeping alarm that indicate when the body is being elevated or retracted shall be furnished. One light shall be located over the front roll up door on the outside and another over the rear door on the outside.

3.5.5.4 Rotating warning light. Two amber rotating lights conforming to MS-51317-4 shall be provided, one at the front, and one at the rear of the body. The lights shall be located above the front and rear doors, but below the roofline of the body. The rotating light shall be operated by means of a separate switch with an indicator light located on the instrument panel and marked to indicate the function of the switch.

3.5.5.5 Lighting and electrical power cable. A main control cable with protective loom shall be furnished and installed between the cab and elevating body to provide all electrical power to the body electrical circuits. Wiring shall be color coded, sized to carry full electrical loads, and shall be in accordance with 3.5.4. Overload circuit protection shall be provided for all lighting and power circuits. The power control cable shall be of sufficient length and so installed and supported to allow full extension and retraction for the full operational range of the body without damage, pinching, or chafing of the electrical insulation.

3.5.5.6 Circuit box. A circuit box shall be furnished. The circuit box shall be provided with a door panel and spring loaded latch. The interior of the door panel shall be provided with a plate showing the lighting schematic. See 3.1.1.1.2 for type of plate.

3.5.6 Hydraulic system.

3.5.6.1 Hydraulic pump. A hydraulic pump with compatible pressure and flow capacities to operate all hydraulic components and accessories simultaneously shall be furnished. The hydraulic pumps shall be driven by a heavy duty power takeoff unit attached to the truck chassis transmission. Split shaft power takeoff is not acceptable. The hoist pump shall be of the gear type design. The pump pressure continuous rating shall be not less than that recommended by the lift platform manufacturer, with an intermittent rating of 500 pounds per square inch (PSI) over the continuous rating. The system hydraulic pressure shall be limited to not more than 2,500 PSI under any operating conditions. The pump system shall include a built in adjustable pressure relief valve(s) or unloading valve(s), with the ranges of adjustment which allow the loads specified herein. The hydraulic pump shall be capable of operating in ambient temperatures of -40°F to +125°F.

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3.5.6.2 Neutral switch. A neutral switch shall be furnished. The neutral switch shall be activated when the transmission is in neutral to operate the pump. A brake lock shall be provided that will not permit the pump to operate unless the brake lock is activated.

3.5.6.3 Automatic throttle actuator. An automatic hydraulic or electric throttle actuator shall be furnished. The actuator shall increase, automatically, the truck engine speed when the pump is being loaded and automatically decrease the speed of the engine when no load is being imposed. The actuator shall interface with the transmission neutral switch (3.5.6.2) so that it is operable only with the transmission in neutral.

3.5.6.4 Hydraulic reservoir. A hydraulic reservoir shall be furnished. The reservoir shall be equipped with a drain, a filler opening cap attached to the reservoir by a corrosion resistant chain and internal baffles to prevent excessive surging of the fluid. A sight gauge shall be provided to observe the level of fluid when the lift body is in its lowered position. A temperature gauge shall be furnished to indicate the temperature of the hydraulic fluid in degrees F. A tank vent shall be furnished and installed to preclude weather and foreign matter entering the reservoir. Provisions shall be made for inspecting and cleaning the reservoir without loss of fluid in the system. The reservoir shall be provided with a photo-etched, corrosion resistant plate installed near the filler cap to indicate the type of hydraulic fluid. The reservoir shall be protected to preclude damage. When specified (see 3.4.1.5.d), the reservoir shall be equipped with a heater.

3.5.6.4.1 Hydraulic fluid. Hydraulic fluid conforming to MIL-H-83282 shall be provided in the hydraulic system.

3.5.6.4.2 Oil filter and strainer. A full flow type 10 micron discharge oil filter with a capacity of not less than three times the pump flow shall be provided in the return line to the reservoir. A 100 mesh suction line strainer with a capacity of not less than two times the pump flow shall be provided in or near the hydraulic reservoir. The return line oil filter shall be equipped with a condition indicator gauge and a 30 to 45 PSI internal bypass. The gauge shall be so located as to be easily readable with the body in the down position. Filter and suction strainer shall be accessible for cleaning or replacement of the filter cartridge.

3.5.6.5 Hydraulic fluid temperature. The hydraulic fluid temperature shall be maintained within safe limits to preclude seal damage and faulty operation of hydraulic valves and controls. Excessive heating of more than 180°F of the fluid shall not be acceptable under any operating conditions.

3.5.6.6 Hydraulic lines and fittings. Hydraulic lines shall be located to facilitate maximum ease of removal and installation, or visual inspection for failures and maximum protection from damage during operation of the body assembly. Flexible hydraulic hose or collector rings shall be used where flexing or hydraulic lines are encountered and where installation of components requires a flexible coupling. Standard straight-thread flared type fittings or adaptions to flared type fittings, or tapered pipe thread with dry seal fits shall be furnished. Hydraulic lines and fittings shall be air and liquid tight and shall be installed to preclude vibration and chafing. High pressure hoses, which are located in areas where chafing may occur, shall be covered with spiral type hose guards. The system shall be designed to preclude hydraulic transients. Provisions shall be made for bleeding air from the hydraulic system and completely draining all hydraulic lines, when necessary. A manual type bleeder shall also be provided as on vehicle equipment on closed centered systems. Hose assemblies and all seamless steel hydraulic lines shall be adequate to provide the rate of flow, temperature and pressures required to operate the system smoothly and efficiently. Galvanized pipe and copper tubing shall not be acceptable. A pressure gauge shall be provided on the discharge side of the hydraulic pump. Accumulators shall not be furnished. The system shall be designed to a safety factor of three for normal operating conditions.

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3.5.6.6.1 Hydraulic valving. An in line relief valve shall be provided, fully adjustable and compatible with the working loads and test load specified herein. The valve shall have provisions for a safety lock wire on the adjustment portion of the valve to permit sealing and locking at the proper working pressure after test. A normally closed 12 volt DC solenoid check valve or valves shall be provided to lock the cylinder(s) in the event of a failure in any other portion of the hydraulic system. If the check valve is not integral with the cylinder, a minimum number of fittings shall be used between the check valve and the cylinder, with a burst strength of at least three times the maximum operating pressure. A normally open 12 volt DC solenoid operated two position two way valve with pilot assist shall be provided to raise and lower the body. The high pressure hydraulic system shall be provided with a gauge port to accomodate a 2 1/2 inch face, 0-3,000 PSI, glycerine filled, stem mounted gauge. The gauge shall be easily visible with the body in the down position. The gauge need not be furnished.

3.5.6.7 Hydraulic cylinders. The lift mechanism shall be actuated by hydraulic cylinder(s). Cylinders may be the multiple telescoping type. All seals, rings, gaskets and other related components shall be designed for easy removal and replacement. Metal nonflexing hydraulic lines shall be used where flexing is not required during operation. An armored or otherwise protected flexible hose shall be used where flexing is required. Provisions shall be made for bleeding the system at all necessary points without removing or loosening any component of the system. The hydraulic system shall be designed to preclude hydraulic transients. Automatic pilot check valves shall be provided to preclude body descent in the event of a hydraulic, electrical or power source failure. A manual device or valve shall be incorporated into the system to lower the body in case of an electrical or power source failure. The device will bleed the fluid back into the reservoir. The device shall provide metering characteristics.

3.5.6.7.1 Hydraulic cylinder valving. A check valve, a return line with filter having antiflowback characteristics, or a ball type shut off valve shall be provided between the reservoir and the return line filter to enable shut off of the fluid supply to facilitate changing the element in the filter. If a ball type shut off valve is used, a caution plate will be installed to warn against operating the pump with the valve closed. A fixed orifice device or an adjustable throttle valve (flow control) shall be located as close as practical to the lift cylinder(s) to limit and control the downspeed. The throttle valve shall be rated for not less than three times the working pressure and shall be equipped with a lock screw or jam nut to hold the desired preset.

3.5.6.8 Controls. Hoist controls shall be electrically operated and shall be of the dead man type. Means shall be provided for manually operating the controls in the event of electrical failure. The controls shall be metered to provide gradual starting and stopping of the hoist and to eliminate shock due to start and stop.

3.5.6.9 Control locations. Control locations shall be as follows: Controls for raising, stopping and lowering the body shall be located at the front end, inside of the body adjacent to the roll up overhead front door (see 3.6.7.4) and inside the vehicle cab. The controls inside the cab shall be located to allow the operator standing on the ground to operate the controls with the driver's side cab door open. Controls shall be capable of overriding each other.

3.5.6.10 Outriggers. Outriggers shall be provided to ensure adequate stability with relation to the ground under the conditions specified in 3.3.3.1 and 3.3.3.7. Outriggers shall be double acting and hydraulically operated. Interlocks shall be provided, allowing the body to raise to the maximum calculated stable height (but not less than 10.0 feet) without outriggers before the outriggers must be extended, to enable the body to raise to its full height. Additional interlocks shall be provided to prevent the transmission from engaging a forward or reverse gear when the outriggers are down. Outrigger controls shall be installed in a readily accessible location adjacent to the lift body controls (see 3.5.6.9), inside the cab to allow the operator standing on the ground, to operate the controls with the driver's side cab door open. Outrigger cylinders, hydraulic fittings, hoses and pilot check valves shall be protected by metal guards and shall be accessible for service. An automatic pilot check valve shall be provided for each outrigger to block the hydraulic fluid in the cylinder and to prevent movement of the outriggers in the event of a damaged hydraulic line.

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Each outrigger shall be provided with an automatic, positive, spring actuated lock or an automatic pilot check valve which will hold the outrigger in the retracted position without settling and which may be manually released. Each outrigger shall be provided with a ground pad. Ground pads shall be of the hinged or fixed type.

3.5.7 Safety. All equipment or exposed portions of the equipment which are subject to extreme temperatures and inclement weather, and all rotating, reciprocating or elevating parts which are of such nature or so located as to become a hazard to operating personnel, shall be insulated, fully enclosed or properly guarded.

3.5.7.1 Safety devices. The following safety devices shall be furnished:

a. A device to prevent sudden lowering of the body through failure of any component of the mechanical, hydraulic, or electrical system other than the hydraulic cylinder(s).
b. A manually operated locking device to retain the body at not less than 8 feet above ground level to the body floor for maintenance purposes.
c. Electrical cut off switch and positive stops to retain the body at any height that the body is located.
d. A device with actuating controls and an indicator installed in the vehicle cab to preclude movement of the vehicle when the body operating system is being actuated and to prevent operation of the body operating system when the vehicle is in motion. The device shall be controlled from the vehicle cab and shall be accessible to the operator when in a seated position. The device shall be installed so that the control cannot be operated unless the detent is placed in the neutral position.

3.5.7.2 Warning signs. A warning sign specifying the load limits for the platform shall be provided on the cargo body, adjacent to the latching side of the forward door on the driver's side. On the inside body adjacent to the front door on the driver's side, a warning sign shall be furnished which reads "KEEP HANDS AND FEET CLEAR OF LIFT WHILE LIFT IS IN OPERATION".

3.5.7.3 Highway warning kit. Three emergency reflective triangles conforming to Federal Motor Carrier Safety Regulation 393.95(f)(2)(i) shall be furnished and stowed in the tool compartment.

3.5.8 Dissimilar metals. All dissimilar metals used throughout the vehicle shall be insulated from one another to preclude galvanic or electrolytic action. Dissimilar metals and methods of protecting them are defined and described in MIL-STD-889.

3.5.9 Lubrication fittings. Lubrication fittings, accessibly located, shall be furnished where moving parts normally require lubrication.

3.6 Body.

3.6.1 Body dimensions. The body shall have the minimum dimensions specified in Table II.

TABLE II DIMENSIONS (Inches)

DESIGNATION	
Cab to Trunnion	100
Length (inside)	168
Height (inside with roll up doors)	78
Width (inside)	90

See Figures 1 and 2 for aircraft dimensions and truck operations. Figure 2 is included for information only and is not a restriction or recommendation on the design of the truck.

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3.6.2 Body. The body shall be provided with a steel frame that is mounted to a steel subframe (see 3.5). The body shall consist of an overhead cab platform with door to the van body interior, and rear doors as specified (see 3.6.7.3).

3.6.3 Body structural strength. Body and structural strength shall be as specified in 3.5.

3.6.4 Body subframing. Body subframing shall be of all steel welded construction, braced at all points of stress. There shall be not less than 12 full width crossmembers, including ends, of full channel or equivalent section, pressed steel construction, attached to longitudinal sills by welds, and at front, rear and every second intervening crossmember by gussets. When subframing is by welded assembly, angle, gussets, crossmembers, longitudinal sills and welded reinforcement (when used) shall be joined by welding for not less than 50% of the length of the contact edge(s). Spacing of crossmember shall be equal, ahead of, and to the rear of the wheels. Longitudinal sills shall be constructed of structural steel channel or structural I-beam or reinforced formed channels. Formed channel sills shall be reinforced within the sill, at each crossmember or body mounting point with formed channel reinforcements. Crossmembers, gussets, formed longitudinal sills and reinforcements shall be not less than 10 gauge (0.1345 inch) steel. Side and front rails (out-rails) shall be not less than 12 gauge (0.1046 inch) steel. The design of the body subframe shall provide tire chain clearance and shall permit vehicle operation at speeds up to 50 MPH over improved roads when loaded with an evenly distributed payload to the rated GVW, without evidence of permanent deformation.

3.6.5 Body framing. Side and front post construction shall be extruded aluminum and shall be not less than 1.26 inches in depth and 0.430 pound per foot. Side and front wall posts shall be spaced on not more than 12 inch centers.

3.6.6 Body interior. The interior of body walls, including front walls and ceiling, shall be lined with not less than 1/4 inch exterior grade plywood. The interior radius corners may be aluminum lined. The understructure and inside wall of the outer skirt shall be covered with a wall coating material to prevent body rusting and body panel drumming. The ceiling shall be insulated with not less than 1 inch fiberglass.

3.6.6.1 Flooring. The floor may be of one piece construction, without welds or seams. Flooring shall be safety aluminum tread plate of not less than 1/4 inch, exclusive of the tread height. When the body floor is constructed of two or more pieces, an automatic seam weld, having full penetration is acceptable without reinforcement. When a seam is hand welded, the underside of the seam shall be reinforced with a 3 inch wide, but not less than, 0.1875 inch thick aluminum butt plate welded between the crossmembers. The floor shall be provided with a covering or a rolled radius around the perimeter of the wall, except at the door openings. The covering or rolled radius shall be the same thickness as the floor and have a vertical height of at least 4 inches and need not have a tread pattern unless the covering and the floor are of the same material. The covering shall be continuously welded when two pieces are furnished. The covering shall be secured with corrosion resistant fasteners to the wall. The floor shall be provided with drains.

3.6.6.2 Interior rope ties. Interior rope ties, either fold down lashing rings, rope cleats or rod types, shall be installed and secured to wall frame members. A minimum of 24 rope ties shall be arranged in three evenly spaced tiers, four rope ties per tier on each side wall; first tier located 5 1/2 inches above the floor line; second tier at 1/3 of wall height; and third tier located at 2/3 of wall height. Rope ties shall not project into the cargo area beyond the wall lining.

3.6.7 Body Exterior. The roof assembly shall be constructed to ensure drainage. Roof support bow construction shall be aluminum and shall be not less than 1.0625 inches in depth and 0.394 pound per foot. Roof bows shall be spaced on not more than 12 inch centers. Roof panel shall be of one piece aluminum construction, not less than 0.040 inch in thickness. The roof panel shall be stretched and secured. The roof panel shall not flutter or "tin can". Body panels shall be not less than 0.051 inch in thickness. Body side panels shall be removable to facilitate body repair. All roof and body seams and joints shall be

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weatherproofed. All body and roof rails shall be extruded aluminum and shall have a maximum radius of 4.5 inches. Rub rails shall be permanently attached to or shall become a part of the exterior of the body at the floor line. Drip moldings shall be furnished over each window at the top, and over the front and rear doors.

3.6.7.1 Windows. Two windows shall be furnished on each side of the body. Window casings, screen, components, parts and accessories shall be constructed of corrosion resistant aluminum. Corrosion resistant properties shall be inherent in the chemical composition of the aluminum. Windows shall be waterproof. Windows shall provide a clear glass opening of not less than 12 inches in width by 18 inches in height. Glass shall be safety plate grade. Windows shall be located midway between the roof and the longitudinal centerline of the body.

3.6.7.2 Window guards. Window guards shall be furnished on the inside of each window. The window guards shall be enclosed in an aluminum frame and hinged to the body. A cam type latch shall be furnished. Screening for the aluminum frame shall be aluminum and shall have an opening of not less than 0.437 inch.

3.6.7.3 Roll up overhead rear door. A roll up overhead rear door shall be provided. The door shall provide a clear opening of not less than 52 inches in width and not less than 72 inches in height. The door shall be plastic covered or aluminum covered 3/4 inch marine plywood construction. Door section corrosion resistant hinges shall be mounted on the inside of the door; not less than four hinges at each section joint. The door track shall be manufacturer's standard corrosion resistant type equipped with a positive stop at the end of the track. Door rollers, a counter balance unit and cables of corrosion resistant construction shall be manufacturer's standard. The door shall be weathertight and shall close not less than 1/4 inch below floor level. A heavy duty corrosion resistant grab handle, closed type, shall be provided on the left outside bottom of the door. One nylon double loop pull down strap, not less than 2 inches wide and 12 inches in length, shall be provided and located adjacent to the door lock. The door shall be capable of being locked from inside and outside the body.

3.6.7.4 Roll up overhead front door. A roll up overhead door at the front of the body load space, centrally located, shall be furnished. The door shall provide a clear opening of not less than 52 inches in width and 72 inches in height. The door shall meet the requirements of 3.6.7.3, except as specified herein. A structural frame member covered with flattened expanded metal shall be installed back of the vehicle chassis cab in alignment with the front door of the body. The expanded metal shall be not less than 16 gauge (0.0598 inch) and shall have an opening of not more than 1 1/2 inches measured across the narrow part of the opening. The perimeter of the expanded metal shall be provided with a molding to cover the raw edges. Frame height shall be up to the chassis cab rear window.

3.6.8 Platform. A platform shall be furnished that is capable of being extended, retracted and moved laterally by hydraulic power. In the retracted position, the front of the platform shall not be forward of the front of the truck. In the raised position, the platform shall extend forward so that the forward edge shall reach at least to the exterior surface of the aircraft at the service doors (see figures 1 and 2), or 7 feet forward of the front of the truck, whichever is greater. These requirements are for clearing the lower fuselage while raising the truck bed. The platform shall have a walkway 48 inches (plus 2 inches, minus 0 inches) wide and shall have a maximum overall width, including guard rails (see 3.6.8.2) of 54 inches. The floor of the platform shall be capable of supporting a 1,000 pound imposed load distributed over any one square foot area. The platform shall be capable of supporting a 750 pound load on the forward end with the platform fully extended and located (a) on the left side with the platform shifted fully left and (b) on the right side with the platform shifted fully right. With the elevating body retracted and resting on its subframe, the platform shall be supported by a structural installation located at the back of the vehicle cab, and the platform shall clear the vehicle cab roof by not less than 3 inches. When the body is being elevated, the upper surface of the platform shall be flush with the van floor. The forward edge of the platform shall have adequate cushioning material to prevent damage to the exterior of the aircraft caused by contact during operation. There shall be a pressure sensing device on the front of the platform that will stop the

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when 5 to 15 pounds of force is encountered. The threshold of the elevating body of the platform shall be provided with means to preclude the feet of operating personnel in the elevating body from being engaged in the platform over the vehicle cab, when the elevating body is ascending. The platform shall remain horizontal as it is extended to its maximum reach. Guard rails shall fasten to and automatically extend, retract and move laterally with the platform.

3.6.8.1 Platform floor. The platform floor shall be one piece, carborundum grit impregnated aluminum safety tread plate not less than 1/4 inch thick, exclusive of tread height.

3.6.8.2 Platform guard rail. A guard rail with a removable front section and telescopic side sections shall be furnished. The guard rail shall be constructed of aluminum structural members and shall be not less than 42 inches in height, measured from the floor to top of the guard rail. The guard rail shall enclose the curbside, streetside and front of the platform. The guard rail shall be designed to move with the platform to cover any extended or retracted position. The front guard rail shall be bifold type on each side of the gangplank. The side guard rails shall be the telescopic type as used by commercial airlines, allowing the aircraft door to be opened while the railing is retracted. Once the aircraft door is open, the railing shall be capable of being readily telescoped to the full forward position. For retraction or extension, side guard rails shall telescope not less than 37 inches. The side guard rails shall be permanently attached. Hangers shall be provided in the body for stowage of the front guard rail.

3.6.8.3 Rear step. A safety tread step shall be provided to the rear of the body to assist personnel, along with the rear bumper (see 3.4.25), into the body interior when the body is in the lowered position. The step shall extend across the width of the body and shall be not less than 10 inches deep.

3.7 Stress analysis. A stress analysis for the body shall be submitted to the engineering office of the procuring activity. The stress analysis shall include the weight of all major items permanently installed on or within the vehicle, and shall show the location of the center of gravity of such components relative to the vehicle axles (raised and lowered positions). In addition, the data shall include the value of stresses in the vehicle chassis frame, its subframe and at the trunnions of the cylinders in the lowest position and when being raised to its highest position. The analysis shall verify that the yield point of the material used is not exceeded under the conditions specified herein. Stress analysis shall be submitted with the vehicle questionnaire.

3.8 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle for operational use including at least the following:

a. Alignment of lights.
b. Adjustment of the engine, electrical and brake systems.
c. Filling and charging of battery.
d. Alignment of front wheels.
e. Inflation of all tires.
f. Complete lubrication of chassis, engine, running gear and mounted equipment with grades of lubricants recommended for the ambient air temperature at the delivery point.
g. Servicing of cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume.
h. Servicing of windshield washer reservoir with water and appropriate additives.

3.9 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted or riveted construction utilized shall be in accordance with the highest standards of the industry.

3.10 First Article. The contractor shall furnish one complete truck for First Article examination and testing to prove that his design and production methods will produce trucks that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 of this specification,

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conducted by the Contractor at his facility or at another Government approved location, and shall be subject to surveillance and approval by the Government. The approved First Article truck shall be rehabilitated to a new condition and delivered as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause of suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First vehicle preliminary inspection.

4.3.1 In process inspection chassis. The first complete truck chassis shall receive an in process inspection at the chassis manufacturer's or body builder's plant by the Government. The inspection shall be conducted to evaluate workmanship and chassis structure conformance to contractual requirements.

4.3.2 In process inspection of body and components. The first complete assembled body and components (subframe, hoist, body, components and assemblies) shall receive an in process inspection at the body builder's plant by the Government prior to prime painting. The inspection shall be conducted to evaluate workmanship and to determine conformance to contractual requirements.

4.3.3 Radio frequency suppression verification. The vehicle manufacturer shall furnish with the questionnaire (commercial vehicle engineering data) a description of all interference suppression devices used to suppress the vehicle electromagnetic radiation in accordance with SAE J551. Data shall be sufficient for visual determination that all required suppression devices are installed on the vehicle.

4.4 First article. The first article shall be inspected and tested by the contractor, in accordance with the approved First Article Inspection Procedure and other requirements specified herein, in the presence of Government representatives. The purpose of the first article inspection and test will be to determine vehicle conformity to the contract.

4.4.1 Weight and payload capacity. The first article vehicle shall be weighed to determine the curb weight and distribution of curb weight on the front and rear axles. It shall then be weighed with a 200 pound load in both the driver's and passenger's seat and the payload specified in 3.2.5. The imposed loading on front and rear axles shall not exceed the suspension, axle, or tire load ratings. The total weight shall not exceed the GVW rating of the truck.

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4.4.2 Road test prior to operational tests. The first article vehicle shall be examined and road tested by the contractor to assure that the vehicle will operate in accordance with contract requirements. The vehicle, loaded with payload and with all equipment specified herein, shall be driven at speeds up to at least 50 MPH, for a minimum distance of 50 miles. The distance traveled shall consist of highway, rough dirt roads, partly surfaced gravel roads, puddles and sharp turns. For each 10 miles of travel, at least one sudden stop shall be accomplished. Upon completion of the road test, all equipment shall be thoroughly examined. The equipment shall show no evidence of mud or dust inside vehicle and there shall be no evidence of racking of the body.

4.4.3 Operational testing.

4.4.3.1 Upsetting and range of movement. With the vehicle resting on approximately level ground and not fastened to any artificial base, the body shall be loaded with twice the specified payload. The body shall be extended to its full height, lowered to travel position and raised to increments of 1/3, 2/3 and its full height, and then lowered to travel position.

4.4.3.2 Wind load test. Wind resistance shall be demonstrated with and without the specified payload. With the specified payload evenly distributed over the body load space, and the body elevated to its maximum specified height measured from the ground, and the vehicle positioned whereby the greatest area is parallel to the center line of the vehicle, the specified wind load, generated by artificial means or by an equivalent simulated wind, shall be applied to the vehicle from the sides. The contractor shall furnish to the engineering office of the procuring activity calculations or a certified test report.

4.4.3.3 Elevating body speed test. With the vehicle resting on approximately level ground and the specified payload uniformly distributed over the body load space, the body shall be elevated to its maximum specified height and held for 2 minutes, plus or minus 1/2 minute, and lowered to rest position. This shall be considered one cycle. Repeat until 40 cycles have been completed. A 30 minute rest period is allowed after 10 cycles if necessary to reduce the hydraulic fluid temperature to an acceptable level. Remove the payload and raise and lower the body to maximum positions at least ten times, continuously. The test shall verify the elevating and retracting speed requirements of 3.3.3.4. The test shall be performed in an ambient temperature of not less than 30°F. The hydraulic fluid shall be stabilized at its operating temperature prior to the performance of this test.

4.4.3.4 Creep test. With the vehicle resting on approximately level ground and an evenly distributed payload (see 3.2.5) placed on the body, the body shall be extended to its full height and held for not less than two hours. Vertical settling shall be not more than as specified in 3.3.3.5.

4.4.3.5 Static conditions. With the body in its most vertically extended position place three times the specified load (see 3.2.5) on the floor of the body for at least 10 minutes. Remove the load and inspect critical areas (see 4.4.3.6) for evidence of cracks or deformation which shall be cause for rejection.

4.4.3.6 Critical areas. At least the following critical areas shall be inspected:

a. Subframe, trunnion, yoke, body cylinder guides, and attaching areas.
b. Struts, hoisting mechanism.
c. Structural steel of body and subframe.
d. Platform support areas.

4.4.3.7 Hoist stability. The hoist shall provide sufficient structural strength to preclude shifting laterally, or fore and aft, without evidence of undue lateral flexing of the structure and without evidence of wear due to binding of the hoist members.

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4.4.3.7.1 Unbalanced load test. With the vehicle resting on level ground, one quarter of the payload specified in 3.2.5 shall be located successively in each corner of the body floor. The hoist shall then raise and lower the body ten times without evidence of binding, deformation or bending of the hoist mechanism. The body with the unbalanced load shall not deflect more than 1-1/2 inches measured from the floor to the top of the subframe.

4.4.3.8 Controls. Controls shall be operated and shall meet or exceed the requirements of 3.5.6.8 and 3.5.6.10.

4.4.3.9 Hydraulic system verification. The hydraulic system shall be verified to assure that the requirements of 3.5.6 through 3.5.6.10 are met.

4.4.3.10 Platform. The platform shall be extended and retracted to ensure the requirements of 3.6.8, 3.6.8.1 and 3.6.8.2 are met. With the floor of the body raised at least 15 feet above ground level, and with the platform fully extended and shifted to full right position, place a 750 pound load on the right side, forward end of the platform for ten minutes. Repeat with the platform fully left and the load on the left side, forward end of the platform. Evidence of cracks or deformation shall be cause for rejection.

4.4.4 Servicing and adjustment of the hydraulic system. Servicing and adjustment of the hydraulic system shall be demonstrated as specified in 3.5.6.

4.4.5 Final road test. The final road test shall be performed after 4.4.3.1 through 4.4.3.10 have been completed. The road test shall be as specified in 4.4.2, except the minimum distance shall be 20 miles.

4.4.5.1 Final operational test. The final operational test shall be performed subsequent to 4.4.4. The body shall be fully elevated and retracted with a uniformly distributed payload of not less than 30 cycles, with a 30 minute interval between each 10 cycles.

4.4.6 Performance test. The body assembly shall be subject to, but not limited to, the following tests:

a. Controls activate when the neutral switch is not activated, when the transmission is in neutral.
b. Controls return to neutral position when the operator releases the control.
c. Controls override each other in the body or in the truck cab.
d. Operate equipment using each set of controls separately.
e. Check to ensure that the entire system can be operated smoothly and without jerking motions in any position.
f. Determine whether the body remains level in all positions with 1 1/2 times the specified loads in the body.
g. Check to determine whether the body can be safely lowered from maximum reach position with rated payload with the hydraulic pump and engine not operating.
h. Check the hydraulic system, including the pump at 10% above the pump manufacturer's maximum operating pressure setting for normal workload conditions.
i. Check limiting devices/switches for function such as aircraft fuselage contact pressure sensing device specified in 3.6.8.

4.5 Production sample. Upon acceptance of the first article vehicle, the vehicle shall remain at the manufacturing facility as a production sample and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.6 Failure. Failure of this first production vehicle to meet the requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under the contract until corrective action has been taken. In addition, subsequent vehicles that do not conform to the accepted production sample vehicle shall be rejected until corrective action has been taken.

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4.7 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting in 3.8 has been accomplished.

5. PACKAGING.

5.1 Vehicle processing. The vehicles shall be processed for shipment from the manufacturer's plant to the initial receiving activity in accordance with the terms of the contract.

6. NOTES.

(This section contains information of general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The vehicles covered by this specification are for nontactical use by the Government in loading and unloading C-5 Aircraft upper deck.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Exterior color other than as specified (see 3.1.1.1).
c. Rustproofing, if required (see 3.1.1.2).
d. Tropical rustproofing, if required (see 3.1.1.2).
e. Coolant, engine oil, battery heater, hydraulic fluid and fuel warmers if required (see 3.4.1.5).

6.3 Performance prediction. Completed copies of SAE Truck Ability Prediction Procedure Form, and computation for low speed requirements for each vehicle model furnished under contract, should be submitted as specified in the contract. The SAE Work Sheet, Item 1, should include vehicle model number, wheelbase and engine model number. Unless other conditions are cited in the contract, computations should be made from normal atmospheric pressure, normal ambient temperature and still dry air. The factors to be used in predicting truck ability (see 3.3.1, 3.3.1.2 and 3.3.1.3) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1	Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	Altitude Factor. 1.00.
Table 3	Rolling Factor. 1.613.
Table 4	Area Factor. 0.173.
Table 5	Velocity Factor. 250.0.
Table 6	Altitude Factor. 1.00.
Table 7	Chassis Friction Horsepower. Use applicable power unit GVW (to nearest, higher 1,000 pounds and the engine RPM to nearest 100 revolutions) which is required for 50 MPH. Geared speed beyond the range of this table factors shall be extrapolated.
Table 8	Grade Factor. 0.75.
Table 8A	Correction Factor. Not required.
Table 9	Road Factor. 0.0.

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6.4 First article. The first article shall be a preproduction model. The contracting officer should include specific instructions in aquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the first article. Approval of the first article shall not relieve the supplier of his obligation to furnish trucks conforming to this specification.

6.5 Subject term (Key Word) listing.

Elevating
High-lift
Material-Handling
Transporter
Van-body

Custodians:
Air Force - 99

Preparing activity:
Air Force - 84

Project No:
2320-F415

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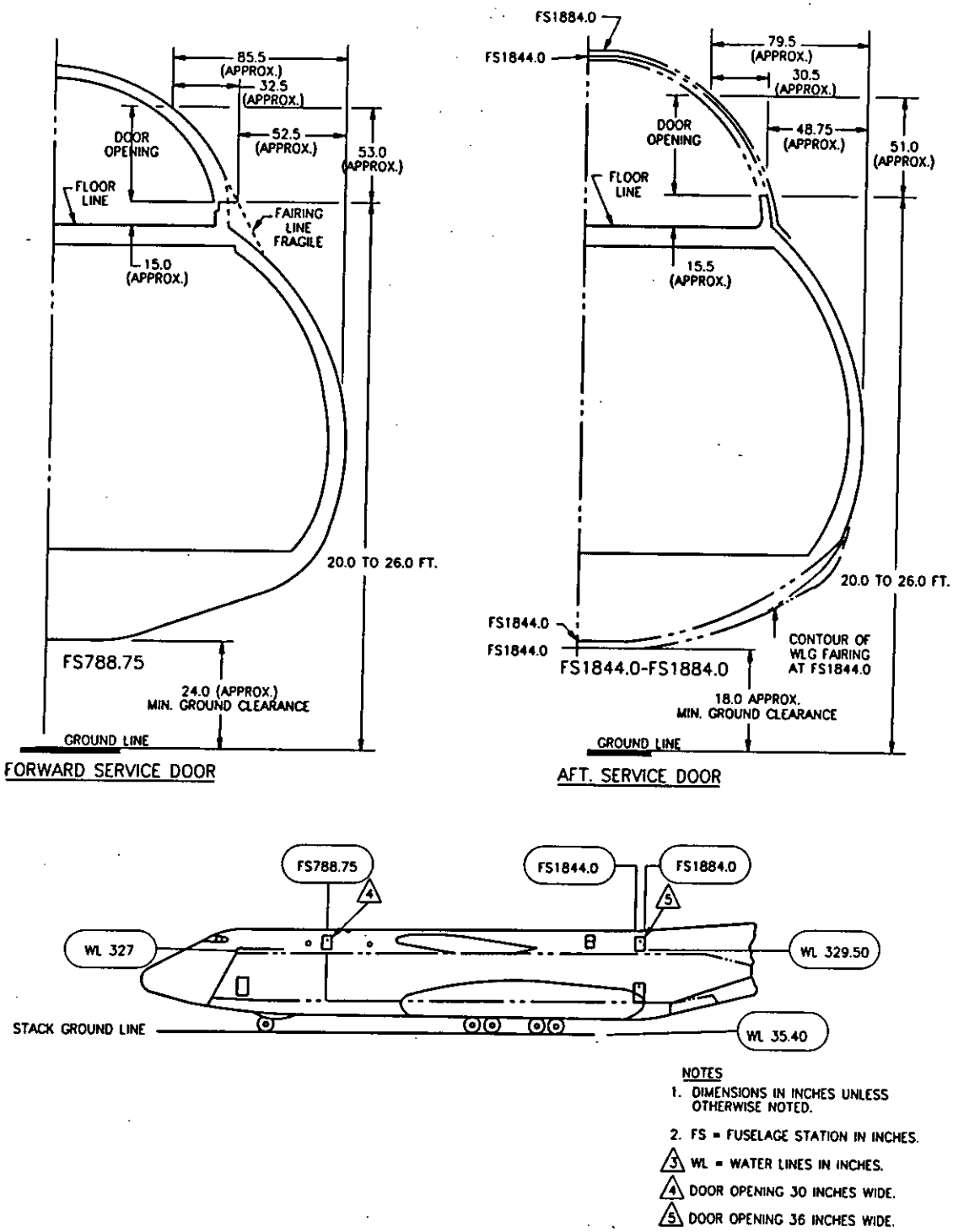


Figure 1
C5 Service Door Parameter

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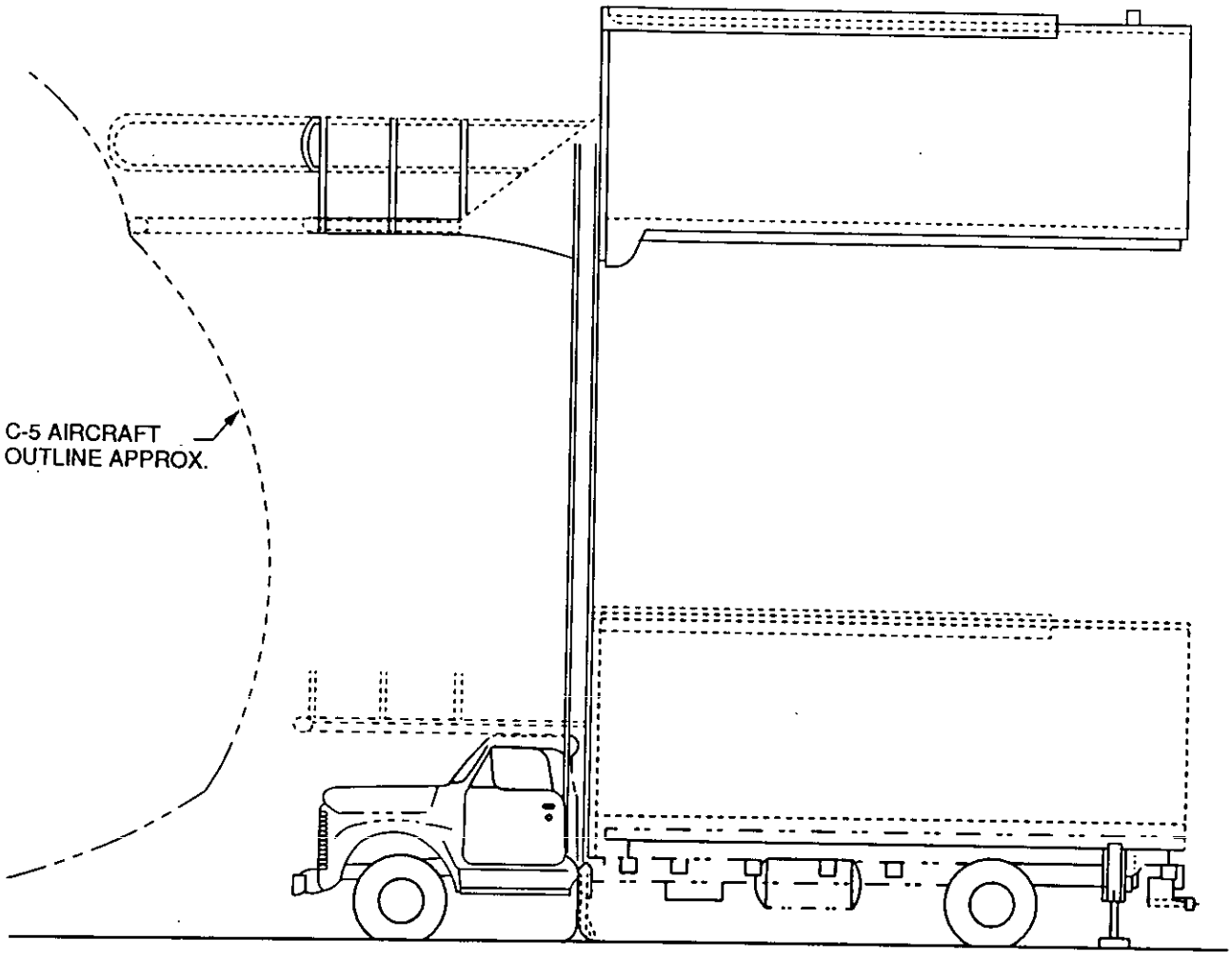


Figure 2

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